

Suhas Sudhir Bharadwaj, Ph.D.

Public Health Researcher / Applied Analytics for Occupational Health & Healthcare Quality

bhara033@gmail.com | (334) 444-4975 | linkedin.com/in/bharadwajss | bhara033.github.io

Education

Ph.D. in Industrial and Systems Engineering at Auburn University, Auburn, AL

Jan 2020 - Aug 2025

Accomplishments

- Data-driven Public Health professional with 5 years of advanced training in Occupational Health and Statistics, applying epidemiologic methods, quantitative analysis, and evidence-based approaches to investigate health risks, evaluate interventions, and develop actionable solutions for complex public health challenges.
- Recognized for excellence in occupational and public health innovation through competitive awards from leading organizations including the American Society of Safety Professionals (ASSP) and National Safety Council (NSC), totaling ~\$22,000, demonstrated community engagement through fundraising efforts supporting the National Multiple Sclerosis Society (NMSS).
- Experienced cross-functional collaborator with demonstrated ability to independently execute research and applied public health projects, including 5 research studies, 10 practice-based projects, 25 graduate credit hours in Statistics, and authorship of 2 peer-reviewed publications in high-impact journals.

Work Experience

Public Health Data Scientist and Mentor

Doctoral Student at Auburn University, Auburn, AL

Jan 2020 - Aug 2025

- Analyzed longitudinal healthcare datasets containing 2.5 million records, 15K entities, and geographical clustering using mixed-effects models to evaluate nursing home quality measures and COVID-19 outcomes, translating statistical findings into evidence supporting healthcare quality improvement.
- Performed large-scale analysis of public health datasets using SAS and R in a high-performance computing (HPC) environment to identify health trends and support occupational health and injury prevention research.
- Conducted longitudinal and compositional data analyses using parametric and non-parametric statistical methods across 135 datasets, 32 participants, and 50+ variables, applying analytical reasoning to support data-driven research findings.
- Showcased collaborative planning and research logistics skills by coordinating approximately 60 field data collection visits, collecting data from 32 manufacturing workers, and facilitating execution by a 5-member team on an NIH-funded research initiative, ensuring efficient and high-quality data collection for downstream analysis.
- Demonstrated strong interpersonal and mentoring skills in engineering education and technical instruction, mentoring 500+ students over 13 semesters, translating complex engineering, programming, statistical, and analytical concepts into accessible instruction and supporting data-driven problem solving.
- Exhibited teaching and communication skills supported by hands-on engineering laboratory instruction, conducting 50+ lab sessions and 2 course lectures to improve student engagement and practical application of course material.
- Optimized student learning and course delivery for 40+ students each semester by streamlining Canvas LMS content management, conducting office hours, providing individualized feedback, and analyzing assignment

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performance, ensuring 100% on-time access to instructional materials and supporting improved learning outcomes.

Research Investigator and Project Leader

Graduate Student at University of Minnesota - Twin Cities, Minneapolis, MN

Sept 2015 - Dec 2017

- Collected, managed, and analyzed behavioral and observational data to evaluate injury prevention interventions and generate evidence supporting transportation safety initiatives.
- Designed and conducted human-subject research, coordinating laboratory and field data collection to generate high-quality physiological and kinematic datasets for statistical analysis, interpretation, and dissemination of research findings.

Additional Experience

Resourcefulness and Leadership in OSH

EHS Specialist at J. M. Huber Corporation, Quincy, FL

Mar 2025 - Jul 2026

- Launched HSMS' LSR initiative, closing all identified compliance gaps for significantly improving safety culture while reducing incident risk as well as site-level plans to achieve 2026 HESS objectives through engaging employees at all levels, building cross-functional teams, and structured action tracking at 2 sites.
- Implemented silica sampling programs across 2 sites, collecting 24 samples according to modified NIOSH 0600/7500 methods and analysing according to modified OSHA ID-142 Gravimetry/X-ray diffraction, resulting in sustained regulatory compliance through silica exposure below 25 µg/m³ (8-hr TWA), improving indoor air quality for 43 employees.
- Established Safety Committees at 2 sites with 20 engaged employees, conducting 20 meetings that generated 10 actionable ideas for health and safety improvements planned for 2026.

Skills

- Statistical Analysis: Regression, Mixed-Effects and Longitudinal Analysis
- Data Management and Reproducible Research: Data Wrangling, Workflow Automation
- Data Visualization and Scientific Communication: Data Storytelling, Dashboards
- Geospatial: Spatial Analysis and Modeling
- Programming and Computational Methods: R, Python, SAS, SQL
- Machine Learning: Feature Engineering, Model Validation
- Public Health and Research Methods: Program Evaluation, Evidence-Based Interventions

Publications

- Ali, H., & **Bharadwaj, S. S.** (2026). Demystifying quality metrics and unveiling the true measure of quality of care in nursing homes. JMIR Human Factors, 13, e72770. <https://doi.org/10.2196/72770>
- Ali, H., Ali, D., Fatemi, Y., & **Bharadwaj, S. S.** (2025). ChatGPT perceptions, experiences, and uses with emphasis on academia. Frontiers in Education, 10. <https://doi.org/10.3389/educ.2025.1559509>

Authorized to work in the U.S. without sponsorship now or in the future